SOIL FACTORS, FARMER'S PRACTICES AND CHESTNUT INK DISEASE: SOME INTERACTIONS

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Abstract

The widespread occurrence of ink disease induced by *Phytophthora cinnamomi* Rands causes important losses in chestnut groves and limits the establishment of new groves in northern Portugal. The existence of both healthy and severely affected chestnut groves in areas where environmental conditions are apparently the same, suggests that farmer's practices, soil factors or their interaction contribute to the incidence of disease. Therefore, eight pairs of adjacent chestnut groves, some severely affected with ink disease and some slightly affected or with no disease, were surveyed in order to determine site characteristics, soil properties and cropping practices. The leaves of chestnuts with different degrees of disease were collected and analyzed for macronutrients. To investigate the relationship between the severity of ink disease and site characteristics, soil properties and cropping practices, data from chestnut groves exhibiting different degrees of infection were submitted to principal components analysis (PCA). The relationship between the severity of ink disease and concentrations of plant nutrients were also submitted to PCA. The importance of soil fertility parameters, soil organic matter and effective soil depth in improving the health conditions of chestnuts was underscored, while radiation, frequency of tillage and imbalanced mineral fertilization contributed to the severity of ink disease. The relative position of the chestnut grove, on the upper or the lower slope affects the hydrological conditions and rooting depth of chestnuts. The results showed that it was not possible to single out one factor as being responsible for the development of chestnut ink disease. Factors that debilitate the trees and reduce their capacity to recover from damage, such as restrictions to root expansion, poor soil fertility, low aeration and soil disturbance by tillage are associated with the occurrence of ink disease.

1. Introduction

Due to the increasing demand for sweet chestnuts (*Castanea sativa*, Mill) farmers are intensifying management practices in order to increase nut production. These include the increasing use of soil cultivation, pruning and mineral fertilizers, which have progressively substituted farmyard manure. However, the current widespread incidence of ink disease induced by *Phytophthora cinnamomi* Rands (Abreu *et al.*, 1993 and Martins *et al.*, 1997) causes important losses in chestnut groves and limits the establishment of new groves. Incidence of ink disease has long been associated with alterations in management practices (Fernandes, 1955; Guerreiro, 1957). Recently, the greedy search for wild mushrooms and their careless collection, in addition to the increased use of tillage, have been damaging the population of ectomycorrhizal fungi in chestnuts groves in NE Portugal, and enhancing the occurrence of ink disease (Marques and Pereira, 1997).

In the Trás-os-Montes region, NE Portugal, a number of chestnut groves severely attacked by ink disease can be found adjacent to areas where the trees are healthy, though climatic and lithological conditions are apparently the same. In healthy groves, it is

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believed that the pathogen exists but produces little or no disease. The absence of the disease in situations normally suited for its development indicates that land management, or an interaction between site factors and cropping practices, might be the cause of disease. There is a lack of empirical work in this subject and the purpose of this study was to investigate possible correlations between soil factors and management practices and the occurrence of ink disease.

Controlling any soilborne root pathogen is difficult and *P. cinnamomi* is no exception. Most efforts made are based on the application of chemicals and the selection of resistant cultivars. However, the greatest hope for control might lie in the integration of the above mentioned approaches with sustainable management practices.

2. Materials and methods

A survey was carried out in adjoining chestnut groves, some showing a severe incidence of ink disease, and others with no or slight disease and all having the same lithological and macro-climatic conditions. In seven pairs the parent material is mica schists and in one case (grove 6) it is granite (Table 1). The soils were classified as anthric Umbrisols and leptic Cambisols, with textures varying from sandy loam to silty loam. Drainage varies from well drained to excessively drained.

Two climatic situations were considered: 1000-1200 mm of mean annual precipitation (sites 1 to 6) and 600-800 mm of mean annual precipitation (sites 7 and 8). Rainfall is the main climatic constraint. Its distribution is erratic with marked dry spells in spring and summer. Temperatures are mild during most of the year (mean temperature is 12°C).

Severity of disease caused by *Phytophthora cinnamomi* was assessed using a 0-4 rating by observation of visual symptoms of ink disease in the tree crown, according to the field keys of Cadahia *et al.* (1991). Management practices were determined by interviewing farmers: tillage was classified according to frequency and use of either draught or tractor power; fertilization practices were scored by considering the amount, frequency and types of nutrients applied. Although pruning is a common practice, it could not be evaluated because farmers' information alone was not sufficient to fully characterize its intensity. So, the views expressed in this paper about pruning are based on systematic observations and research published by Pires and Portela (1993, 1997), rather than on interviews with farmers. The complete description of management practices was published by Portela *et al.* (1997). Site characterization of the 16 chestnut groves was described for landform and topography, and three to five soil pedons in each grove were described according to FAO-ISRIC (1990). In order to compare soil fertility parameters, a higher density of sampling was carried out. The average number of samples collected in each chestnut pair was 22 at two depths (0-20 and 20-40 cm). In the affected groves soil samples were collected under trees whith a greater incidence of disease. To evaluate nutrient status of chestnuts, leaves were collected from 72 trees with several degrees of infection. The sampling procedures and the macronutrients analysis methodology were as previously described (Portela *et al.*, 1997).

To investigate the relationship between the severity of ink disease and site characteristics, soil properties and cropping practices, data from chestnut groves exhibiting different degrees of infection were subjected to principal components analysis (PCA). The data were organized in a matrix, and Pearson's correlation coefficients were determined, in order to decide which were the important characteristics to consider in the PCA. Only the relevant and more quantifiable variables (Table 4) were selected to carry out PCA, using the SYSTAT statistical software. Since manuring and liming are revealed in soil fertility parameters, and because they were carried out on an irregular basis and in only a few cases, they were excluded from the PCA, to avoid redundancy. The relationship between the severity of ink disease and concentrations of plant nutrients were also submitted to PCA.

3. Results and discussion

Some relevant site characteristics and soil profile descriptions of the 16 chestnut groves are summarized in Table 1 and soil fertility parameters are indicated in Table 2. Individual Pearson's correlation coefficients between degree of infection and management practices, site characteristics and soil properties were calculated, the significant ones ($P<0.05$) being indicated in Table 3.

As shown in Table 1 severely affected chestnut groves are more frequently found in south facing slopes. Greater exposure to sunlight means higher soil temperatures, higher soil organic matter decomposition and favourable conditions for a water deficit in the summer. So, radiation is the only site characteristic that appears to be relevant in predisposing chestnut trees to ink disease. On the contrary, ink disease decreases with deeper rooting and with better soil fertility conditions: organic matter, extractable phosphorus and exchangeable calcium and potassium. As shown in Table 2, the lower soil fertility level of the affected groves is indisputable, although it is not possible to point out among the soils studied a unique soil chemical parameter that allows us to make a clear-cut distinction between the soils under more healthy or diseased chestnuts.

As far as management practices are concerned, data indicate that severity of disease is positively correlated with the intensity of tillage operations and fertilization; and it is negatively correlated with the application of manure and liming (Table 3).

PCA (Table 4) shows that characteristics related to tree nutrition were the most important variables associated with the first PC. The influence of tillage on the incidence of disease is underscored by PC 2; PC 3 seems to be associated with some outsiders, where fertilization presents a strong negative correlation with ink disease, in opposition to PC 1. So, the importance of soil fertility parameters and rooting depth in improving the health conditions of chestnuts was shown, while radiation and mineral fertilization (mostly N) contributed to severity of ink disease. Unexpectedly, fertilization is in opposite relation to nutrient status in the soil. Interviews with farmers revealed that healthier groves received a lower nitrogen application than affected ones. Since N is a growth stimulating nutrient, the content of the other nutrients in the soil might have been depleted by an increased demand from the tree. However, if nutrient applications were balanced, fertilization might be used to improve nutrient status and health condition of the trees, as shown by the third PC. Graphical analysis of this PC shows that fertilization in an equilibrated basis contributes to the good health conditions of chestnut groves, in sites 3H, 4H and 8H.

The soil fertility indices that showed higher correlation with good health conditions were exchangeable bases, K, Ca and Mg followed by extractable P. Rooting depth is the physical parameter that most affects tree nutrition, since it determines the soil volume contributing to nutrient exploitation. Shallow soils are usually poorly buffered against external extremes, e.g. root damage by tillage and drought periods. It is worth noticing that radiation is also associated with tree nutrition. It negatively affects the health conditions of chestnut trees, which we ascribe to increased water deficit in spring and summer. Pearson's correlation coefficients between degree of crown and soil nutrient indices show that Ca and K are the nutrients with higher correlation coefficients ($P<0.001$). Pearson's correlation coefficients between degree of symptoms and leaf nutrient concentrations (data not shown) corroborate the soil data, Ca and K being again the nutrients with the highest correlation coefficients ($P<0.001$).

The importance of soil fertility indices in health conditions of chestnuts are also shown by the PCA performed for nutrient content in leaves (Table 5), with an accumulated variance of the three PCs accounting for nearly 90% of variation. According to the first PC, nitrogen is not relevant for quantifying the disease, P, K Ca and Mg being the other macronutrients, strongly negatively associated with the disease. However, as shown by PC2, nitrogen is strongly negatively associated with the disease. These results are in agreement with the data indicated in Table 4 for the PC3, where balanced fertilization

(NPK) may improve the health conditions of chestnuts. The third PC is not discussed here because this vector is associated with some outsiders.

Leaf analysis (Table 6) shows that Ca undergoes the highest reduction in affected trees, when compared with the other nutrients. The role of K and Ca in improving the resistance of plants to root diseases has long been recognized, whilst N increases the susceptibility of the host to fungal parasitic diseases (Kiraly, 1976; Huber and Arny, 1985; Ko and Kao, 1989). Susceptibility of plants to infection has been related to low Ca content in tissues. Calcium is required in middle lamella for stability of biomembranes (Marschner, 1986). The activity of the pectolytic enzyme produced by parasitic fungi, which dissolves the middle lamella, is drastically inhibited by calcium (Marschner, 1986; Ko and Kao, 1989). In addition, Ca plays a major role in root growth, which is obviously crucial for the plant to regenerate new roots. Shea and Broadbent (1983) report several studies that show the positive effect of Ca in the control of diseases caused by *Phytophthora.* As far as N is concerned, it is generally believed that the severity of disease caused by root infection fungi increases when large amounts of nitrogen are applied to the soil (Kiraly, 1976; Henis, 1976; Marschner, 1986), because excess N (imbalance) may produce thin cell walls, thus allowing fungi to penetrate more easily. As mentioned above liming had a positive influence on tree health, which may have improved the calcium nutrition of chestnuts.

The opposite variation of soil organic matter and depth of rooting against the number of tillage operations is shown by the second PC (Table 4). Soil disturbance increases soil organic matter mineralization so, healthy or slightly affected chestnut groves have higher organic matter content (Table 2). Farmyard manure and other organic dressings were applied to some of the healthier chestnut groves in sites 1, 3, 4, 5 and 6. Application of organic materials to the soil has long been recognized as a procedure that protects hosts from soilborne diseases (Malajczuk, 1979; Shea and Broadbent, 1983; Cook, 1984). On the one hand it improves the soil conditions for the host survival, and on the other hand it provides a means for biological control of the pathogen.

While manuring improves soil organic matter, heavy pruning reduces the amount of literfall incorporated into the soil and also increases soil exposure to sunlight. Defoliation and crown transparency characterize the chestnut groves affected by the ink-disease and expose the soil surface to sunlight (Pires *et al.*, 1997; Pires and Portela, 1997). In most healthy chestnut groves there are dense canopies that lower soil temperatures, while in the affected ones the large open areas enhance organic matter decomposition. Pruning and removal of the total pruned material removes remarkable amounts of calcium from chestnut groves, which may contribute to the impoverishment of Ca in the soil. Nutrient cycling studies in chestnut conducted in this area show that in frequently pruned chestnut groves lower amounts of litterfall are incorporated into the soil (Pires and Portela, 1993, 1997), therefore they tend to be poor in organic matter.

Concerning rooting depth two different situations must be considered: shallow bedrock associated with convex slope forms or strongly sloping land and the presence of a compacted layer in concave land. In sloping land with a very shallow bedrock, soil disturbance by tillage frequently had disastrous effects, due to erosion: it increases soil loss in convex areas and deposition in concave areas. Soil disturbance, besides accelerating organic matter decomposition, also reduces structure stability, hence enhancing soil erosion. The reduction of soil aggregation due to tillage promotes crust formation under the beating action of rain drops. When this process occurs, farmers tend to till the soil after each rainfall in order to break up crusts and to increase the infiltration rate. However, the schist-derived soils have a poor base and low organic matter content, therefore easily allowing the formation of crusts which greatly impede water infiltration. The effect of tillage on infiltration seems to be very transient, because the impact of rain compacts the soil rapidly and the rates of water infiltration are soon reduced to their former level. Therefore run-off increases. Transportation of fine fractions causes surface sealing in low land and water ponding on the soil surface during rainy periods. In three affected chestnut groves (2A, 4A, and 5A) the shallow rooting depth is due to a plough

pan at about 25 cm depth rather than to superficial bed-rock. Although the influence of soil texture on the degree of infection was not shown by the statistical analysis, in these three cases, where chestnut root development was restricted by such a layer, the amount of fine sand plus silt fractions is about 70%. The footslope position favours the accumulation of fine sand and silt, which in addition to frequent tilt favours compaction of the subsurface horizon. This position also contributes to inefficient vertical drainage and/or low oxygen availability during short, wetter periods, although no evidence of poor drainage was observed in any grove. However, even for short periods, low oxygen availability is a critical factor that favours Phytophthora root disease (Shea and Broadbent, 1983), particularly in the spring. Reduced rooting depth and probably low oxygen availability decreases the ability of trees to regenerate roots. Trees that have deeper root systems are not so dependent on surface feeder roots for the uptake of nutrients and water. Even if roots are damaged by tillage and/or affected by fungus they have a higher regenerating capability.

Soil cultivation has a systematic destructive effect both on the superficial roots and on the network of mycorrhizal mycelium (Marques and Pereira, 1997). Some Basidiomycetes that grow under the chestnut trees in this study area produce ectomycorrhizae and the protective effect of mycorrhizal symbiosis against ink disease has already been referred to by Marques and Pereira (1997). These authors observed that the abundance of mushrooms is higher in non-disturbed groves. Root wounding creates favourable conditions to fungus invasion and spread; it also devitalizes the trees dramatically, since feeder roots are usually most abundant closer to the soil surface.

In conclusion, we cannot single out one factor responsible for development of chestnut ink-disease, because of the many interactions between biological, chemical and physical soil properties, which influence both the susceptibility of chestnut, or, the success of the pathogen. However, the results indicate that certain circumstances promote the severity of symptoms, such as site and soil physical conditions, and management practices. South facing slopes have higher radiation, lower organic matter content and are more prone to erosion. Low effective soil depth, both due to shallow rock or promoted by frequent soil cultivation reduces rooting volume. Deep and expanded rooting and higher soil fertility conditions facilitate better nutrition and storage of reserves, which may serve as a substitute support system during the temporary reduction in nutrient uptake after a fungus attack. Cropping practices that contribute to the soil organic matter impoverishment (due to increasing rate of mineralization or low incorporation of litterfall or organic residues) create less favourable conditions to chestnut growth, predisposing the trees to severe damage. In addition, management practices that cause biological disturbance, such as breakdown of mycorrhizal mycelium and root damage seem to act directly in promoting disease severity.

Since the control of ink disease is difficult, an appropriate management of the interaction soil/plant/microbial by means of cultural practices might be a promising approach for achieving long term productive chestnut groves.

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Table 1. Degree of crown symptoms, some site characteristics and soil profile properties of the groves

Groves	Degree of crown symptoms	Slope form	Slope %	Slope aspect	Rooting depth (cm)	Surface stoniness (%)	Fine sand +silt (%)
1H Padrela	0.4	Straight	5-15	NE	60-100	40-60	61
1A	1.7	Straight	5-15	NE	30-80 [a]	60-80	58
2H Seixedo	0.5	Ccav	5-10	NE	60-100	40-60	74
2A	1.9	Ccav-Cvex	0-10	S	20-60 [ab]	60-80	73
3H Redondelo	0.0	Ccav	5-10	E	>100	20-40	65
3A	2.4	Cvex	10-20	SE	40-80 [a]	>80	63
4H Fonte-Vide	0.5	Ccav	3-15	S	>100	20-40	69
4A	3.5	Ccav-Cvex	5-20	E	25-100 [b]	40-60	68
5H Carrazedo	0.7	Straight	5-10	NE	40-50 [a]	40-60	74
5A	2.6	Ccav	2-5	E	25-100 [b]	20-40	73
6H Argemil	0.9	Ccav	2-5	S	65-80	40-60	43
6A	2.6	Straight	10-15	SE	30-85 [a]	40-60	43
7H Espinhoso	0.6	Ccav	10-15	N	>100	40-60	55
7A	2.1	Cvex	2-5	W	40-100 [a]	60-80	56
8H Corujas	0.1	Ccav	2-5	E	60-100	40-60	59
8A	2.1	Cvex	5-15	S	25-40 [a]	>80	61

H- healthy or slighted affected; A- affected; Ccav- concave slope; Cvex- convex slope; a- shallow rooting due to hard rock; b- shallow rooting due to a plough pan.

Table 2. Soil fertility parameters in chestnut groves: healthy or slightly diseased, and affected with ink disease. Average values of the eight pairs

	Healthy or slightly affected groves		Affected groves	
Depth (cm)	00-20	20-40	00-20	20-40
pH $(H_2O)^+$	4.9 a	4.9 b	4.9 a	4.7 b
pH $(KCl)^+$	4.0 a	4.0 b	4.0 a	4.0 b
Organic matter (%)	2.3 a	1.7 B	1.6 a	1.0 B*
Extractable P_2O_5 (mg kg^{-1})	76 A	37 B	55 A*	13 B*
Exchangeable cations ($cmol_c$ kg^{-1})				
Ca^+	1.23 A	0.68 b	0.78 A*	0.64 b
Mg	0.41 a	0.27 b	0.33 a	0.28 b
K	0.42 A	0.30 B	0.33 A*	0.19 B*
H + Al	1.3 a	1.6 B	1.5 a	2.0 B*
Base saturation (%)	62 A	46 B	51 A*	36 B*

Organic mater by the modified Walkley-Black method; pH- 1:2.5 soil solution ratio; P_2O_5- Egner-Riehm method; Exchangeable bases- 1N ammonium acetate pH 7; H+Al- KCl method. $^+$- grove 2A not included in the mean, since soil was limed 2 years before sample collection, when the disease was already established. Means within the same row and the same depth followed by the same lowcap letter are not significantly different; and when followed by the same capital letter are significant by Student's t test (* - $P<0.05$).

Table 3. Significant Pearson's correlation coefficients ($P<0.05$) between the degree of infection and management practices, site characteristics and soil properties

	Till	Fert	Man	Lim	Rad*	RD	OM	P	Ca	K
Disease	0.541	0.273	- 0.405	- 0.392	0.399	- 0.606	- 0.331	- 0.473	- 0.514	- 0.589

Till- tillage operations; Fert- fertilization; Man- manuring; Lim- liming; Rad - radiation; RD- rooting depth; OM- organic matter; P- extractable phosphorus; Ca- exchangeable calcium; K- exchangeable potassium.
* - Radiation was obtained combining the slope gradient with slope aspect following the procedure of Rodrigues (1988).

Table 4. Principal component analysis in chestnut groves with several degrees of ink disease

	Eigenvectors and correlation					
	PC1	R	PC2	R	PC3	R
Eigen value	3.205	-	1.749	-	1.243	-
Accumulated variance (%)	35.18	-	54.37	-	68.01	-
Variables						
Tillage	- 0.063	- 0.204	0.375	0.655	- 0.179	- 0.223
Fertilization	- 0.160	- 0.514	0.096	0.168	0.616	0.766
Radiation	- 0.223	- 0.715	- 0.008	- 0.014	0.318	0.395
Rooting depth	0.164	0.526	- 0.390	- 0.682	0.040	0.049
Organic matter	0.059	0.188	- 0.384	- 0.671	0.281	0.349
Phosphorus	0.168	0.538	0.047	0.082	- 0.173	- 0.215
Calcium	0.253	0.810	0.176	0.300	0.276	0.343
Magnesium*	0.186	0.596	0.295	0.516	0.288	0.358
Potassium	0.275	0.880	0.065	0.114	0.149	0.186
Correlation between disease and eigen vectors	-	- 0.700	-	0.382	-	- 0.126

* Although the Pearson's correlation coefficient for magnesium was not significant (R= - 0.191) this nutrient was considered in the PCA because base saturation could be substituted by the exchangeable bases Ca, Mg and K.

Table 5. Examination of principal components of the nutrient content in leaves of 72 trees collected in chestnut groves with several degrees of ink disease

	Eigenvectors and correlation					
	PC1	R	PC2	R	PC3	R
Eigen value	2.46	-	1.12	-	0.92	-
Accumulated Variance, %	49.23	-	71.60	-	89.90	-
Variables						
N	0.075	0.185	0.713	0.797	0.615	0.563
P	0.327	0.804	0.241	0.270	- 0.314	- 0.287
Ca	0.352	0.867	- 0.155	- 0.173	0.385	0.352
Mg	0.301	0.742	- 0.491	- 0.550	0.306	0.280
K	0.281	0.692	0.250	0.280	- 0.611	- 0.559
Correlation between disease and eigen vectors	-	- 0.548	-	- 0.247	-	- 0.097

Table 6. Mean nutrient concentration of leaves of healthy chestnuts and inaffected trees

	Healthy groves	Affected groves	
Degree of disease	0	0-1	2-3
Nutrient			
N, g kg-1	18.77 aA	18.53 aB	14.31 bA**B**
P, g kg-1	1.93 aA	1.72 ab	1.43 bA**
K, g kg-1	11.49 aA	10.61 aB	8.52 bA***B*
Ca, g kg-1	5.31 aA	4.74 aB	3.37 bA***B*
Mg, g kg-1	1.63 aA	1.27 ab	1.19 bA*

Means within the same row followed by the same lowcap letter are not significantly different, and when followed by the same capital letter are significant by Student's t test (* - $P^20.05$; ** - $P^20.01$; *** - $P^20.001$).